

Advay Kadam

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science and Statistics, Minor in Mathematics | GPA: 3.9

May 2027

- **Relevant Courses:** Machine Learning, Distributed Systems, System Programming, Deep Generative Models, Time Series Analysis, Algorithms and Models of Computation, Data Structures and Algorithms

EXPERIENCE

Capital One

Oct 2025 – Present

Incoming Software Engineering Intern

Richmond, VA

- Summer 2026

Bank of New York Mellon

Jun 2025 – Aug 2025

Data Science and Engineering Intern

Jersey City, NJ

- Integrated RAG based agent to automate metadata security classification across **4,000+** datasets in data lake
- Automated dataset tagging via chain of thought reasoning agent, reducing manual audit time by **15 hours/week**
- Reduced end-to-end latency by **83%** for multi-LLM-agent pipeline through multithreaded input processing
- Enhanced metadata-driven quality checks with RAG agent, boosting detection of low-quality metadata by **35%**

Procter & Gamble

May 2024 – Aug 2024

AI Engineering Intern

Cincinnati, OH

- Cut sales ETL pipeline refresh time by **60%** on **200k+** records via optimized SQL and PySpark transformations
- Accelerated sales dashboard update speed by **30%**, improving responsiveness for **300+** analysts
- Implemented statistical anomaly detection using Isolation Forests with Python to flag deviations in batch loads

RESEARCH AND CAMPUS INVOLVEMENT

Supercomputing System and AI Lab @ UIUC

Aug 2025 – Present

Machine Learning Undergraduate Researcher

Urbana, IL

- Researching how to enable efficient quantized automated compression for MoE (Mixture of Experts) models
- Analyzing quantization techniques to compress LLMs, benchmarking performance across accuracy and memory
- Tested VLM QA inference task using efficient quantized inference on AWS Trainium and Delta Supercomputer

Siebel Center of Computing and Data Science

Jan 2025 – Present

Course Assistant for Data Structures and Algorithms

Urbana, IL

- Leading office hours and discussion sessions on abstract data types, C++ algorithms, and complexity analysis

National Center for Supercomputing Applications @ UIUC

Aug 2023 – May 2024

Undergraduate Researcher

Urbana, IL

- Analyzed **2+** PB of real-time network metadata using Pandas, NumPy, Shell scripting to assess PQC resilience
- Developed Flask dashboard to parse Zeek logs and deliver live metrics on PQC algorithm usage and vulnerability

PROJECTS

Hybrid Distributed File System (HyDFS) | C++, Sockets (UDP/TCP), Synchronization

- Built Cassandra inspired distributed file system with consistent hashing and 3× replication across 10 machines
- Optimized replication and merge for eventual consistency, improving throughput 2.7× and reducing overhead 35%
- Integrated a UDP-based gossip heartbeating protocol for decentralized node liveness detection and fault tolerance

Custom Memory Allocator | C

- Designed and optimized a low-level memory allocator from scratch, outperforming GNU libc allocator by 14%
- Leveraged custom heap checker to test allocator coalescing, free-list optimization, and cache-aligned metadata

IMC Prosperity (International Trading Competition) | Python, Pandas, Matplotlib, Seaborn

- Placed in the top **3%** in a 15-day global trading challenge, writing Python scripts for algorithmic trading
- Applied Backtesting and implemented strategies related to market making, Black-Scholes pricing, and regression

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, R, Java, HTML/CSS

Tools: PyTorch, TensorFlow, NumPy, Pandas, Flask, AWS SageMaker, Apache Spark/PySpark, Matplotlib, Git

Awards | **3rd place** at UIUC Spring 2024 Datathon, **2nd place** at UIUC Fall 2023 Datathon, **2x** Dean's List